



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Create RESTful APIs and Web Apps with Python Flask

TGS Ref No: TGS-2021010365

Conducted by

Tertiary Infotech Academy Pte Ltd

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Version 11.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
10.0	Prior release	Legacy Flask lesson plan retained as the content-coverage reference.	Dr Alfred Ang
11.0	5 September 2026	Rebuilt as a two-day FastAPI/OpenAPI 3.1 course with 12 Northstar Commerce labs and aligned assessment.	Dr Alfred Ang

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Course Information

Course	Create RESTful APIs and Web Apps with Python Flask
Course code	TGS-2021010365
TSC	Applications Integration (ICT-DIT-3003-1.1)
Duration	2 days · 16 training hours
Assessment	WA 60 min + PP 90 min

Learning Outcomes

- LO1: Identify and assess FastAPI as middleware for creating connections between web clients, applications and data stores.
- LO2: Integrate data and functions into a browser web app through typed FastAPI routes and reusable Python services.
- LO3: Support API-level integration using REST semantics, JSON and an OpenAPI 3.1 contract.
- LO4: Perform tests and checks on FastAPI-to-SQLite connections using Swagger UI, pytest, logs and VS Code debugging.
- LO5: Modify a FastAPI service to improve integration, validation, compatibility, performance and security.

Slide and Lab Alignment

Topic	Slides	Labs	Codes
01 REST API Fundamentals and AI Vibe Coding	22–52	1, 2	K1, K2, A1, A2
02 AI-Assisted API Functions, Routing and Security	53–81	3, 4	K2, K4, A3, A8
03 Request Handling, Data Validation and JSON Responses	82–110	5, 6	K2, K5, A3, A4
04 API Error Handling, Testing and Debugging	111–139	7, 8	K4, A5, A6, A7, A8
05 API Architecture, Data Models and Database Integration	140–168	9, 10	K3, K4, A3, A5, A6
06 REST API Documentation, Integration and Deployment	169–197	11, 12	K3, K5, A4, A7, A8

Day 1 Schedule

Time	Duration	Facilitation and Content	Slide/Lab Reference
09:00–09:30	30 min	Administration, outcomes and course contract	Admin/review slides
09:30–11:00	90 min	Topic 1 concepts and demonstrations	Slides 22–52
11:00–12:30	90 min	Labs 1-2	Lab 1 · slides 41–46; Lab 2 · slides 47–52
12:30–13:30	60 min	Lunch	Admin/review slides
13:30–15:00	90 min	Topic 2 concepts and demonstrations	Slides 53–81
15:00–16:15	75 min	Labs 3-4	Lab 3 · slides 70–75; Lab 4 · slides 76–81
16:15–17:15	60 min	Topic 3 concepts and demonstrations	Slides 82–110
17:15–18:00	45 min	Labs 5-6 briefing and guided start	Lab 5 · slides 99–104; Lab 6 · slides 105–110

Instructional time totals 8 hours including facilitated lab practice; lunch is excluded.

Day 2 Schedule

Time	Duration	Facilitation and Content	Slide/Lab Reference
09:00–10:30	90 min	Topic 4 concepts and demonstrations	Slides 111–139
10:30–12:00	90 min	Labs 7-8	Lab 7 · slides 128–133; Lab 8 · slides 134–139
12:00–13:00	60 min	Topic 5 architecture and SQLite	Slides 140–168
13:00–14:00	60 min	Lunch	Admin/review slides
14:00–15:30	90 min	Labs 9-10	Lab 1 · slides 41–46; Lab 9 · slides 157–162; Lab 10 · slides 163–168
15:30–16:30	60 min	Topic 6 integration and deployment	Slides 169–197
16:30–17:45	75 min	Labs 11-12 capstone	Lab 1 · slides 41–46; Lab 2 · slides 47–52; Lab 11 · slides 186–191; Lab 12 · slides 192–197
17:45–18:00	15 min	Review, TRAQOM and assessment briefing	Admin/review slides

Instructional time totals 8 hours including facilitated lab practice; lunch is excluded.

Assessment Plan

Written Assessment (SAQ): 10 open-ended questions, K1-K5, 60 minutes, open book.

Practical Performance: 6 products, A1-A8, 90 minutes, open book.

The Practical Performance tasks use the same prompt-contract, FastAPI, OpenAPI, SQLite, testing and commerce-web-app evidence practised in Labs 1–12. AI-generated drafts are accepted only after human review and executable evidence.